5.2 Congruent Polygons

Name _____ Period ____ Date _____

Geometry 5.2 (30-minute plan) · Student sheet

I can…

- identify corresponding parts of congruent figures.
- use corresponding parts to find unknown values.
- use the Third Angles Theorem.

Vocabulary · fill in the missing word

Corresponding parts — a pair of sides or angles that have the same relative position in two _____ figures

Congruent figures — figures that have the same size and _____

Part 1 · Notes and examples (fill in each note, then work the example)

Congruence statements: in △ABC ≅ △DEF, the order of the letters tells which parts _____.

Example 1

△ABC ≅ △DEF. List all pairs of congruent corresponding parts.

Corresponding parts of congruent figures are _____.

Example 2

△ABC ≅ △DEF. AB = 2x + 3 and DE = 11. m∠C = (5y + 10)° and m∠F = 60°. Find x and y.

Third Angles Theorem: if two angles of one triangle are congruent to two angles of another triangle, then the third angles are _____.

Example 3

In △ABC and △DEF, m∠A = m∠D = 45° and m∠B = m∠E = 70°. Find m∠C and m∠F. What can you conclude?

Part 2 · Partner practice

Partner practice

△MNP ≅ △RST. MN = 3x − 4 and RS = 17. Find x.